



The pure-quartic soliton laser

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Ultrashort pulse generation hinges on the careful management of dispersion. Traditionally, this has exclusively involved second-order dispersion, with higher-order dispersion treated as a nuisance to be minimized. Here, we show that this higher-order dispersion can be strategically leveraged to access an uncharted regime of ultrafast laser operation. In particular, our mode-locked laser—with an intracavity spectral pulse shaper—emits pure-quartic soliton pulses that arise from the interaction of fourth-order dispersion and the Kerr nonlinearity. Phase-resolved measurements demonstrate that their pulse energy scales with the third power of the inverse pulse duration. This implies a strong increase in the energy of short pure-quartic solitons compared with conventional solitons, for which the energy scales as the inverse of the pulse duration. These results not only demonstrate a novel approach to ultrafast lasers, but more fundamentally, they clarify the use of higher-order dispersion for optical pulse control, enabling innovations in nonlinear optics and its applications.

Ultrashort lasers have been fundamental for the development of major photonic applications such as telecommunications^{1–3}, supercontinuum^{4,5}, nonlinear imaging⁶ and frequency comb generation⁷. Soliton effects—which are based on the balance of quadratic dispersion and nonlinearity—have allowed for the direct generation of optical pulses with durations below 10 fs (ref. ⁸). Soliton fibre lasers, which have relatively simple configurations, can be built using off-the-shelf components^{9–11}; however, the energy of the pulses that is emitted by these lasers is limited by the soliton area theorem^{12,13}, a fixed relation between the energy and the width of the soliton pulse, and by the appearance of resonant spectral sidebands that arise from periodic perturbations in the optical cavity¹⁴. More complex ultrafast laser configurations were developed to overcome these limitations^{15,16}, which emit highly chirped pulses that require external recompression, reducing the efficiency. Although this level of complexity may not be a strong impediment for advanced applications that require exceedingly high energies, many other applications could benefit from the simplicity and efficiency of soliton lasers that build on the shape-preserving, transform-limited properties that are inherent to soliton pulses; however, new fundamental scaling laws are required to overcome the limitations of soliton lasers so far.

A series of studies in the early 1990s proved that the minimum pulse duration in ultrafast soliton lasers is limited by higher-order dispersion^{17,18}. Third-order dispersion leads to pulse instabilities and energy loss through dispersive wave emission^{19,20}. Nonetheless, the existence of stable soliton-like pulses in the presence of both negative second- and fourth-order dispersion (FOD) has been theoretically predicted^{21–24}.

Pure-quartic solitons (PQSs)—shape-maintaining pulses that arise from the balance of the Kerr nonlinearity and negative FOD—were recently discovered in dispersion-engineered photonic crystal waveguides²⁵. They were observed in a spectral range where the second-order (quadratic) dispersion β_2 was positive (normal), the third-order (cubic) dispersion β_3 was negligible and the fourth-order (quartic) dispersion β_4 was negative (anomalous). Recent theoretical studies have unveiled that PQSs possess advantageous energy scaling^{26,27} that grants them the potential to achieve substantially

higher energy than their conventional soliton counterpart for short pulse durations. This discovery has led to efforts to transition from planar silicon photonic crystals to other platforms^{26,28}, including optical fibres where mature fibre laser technology can be utilized. Initial approaches involved carefully designed photonic crystal fibres with multilayer air holes that exhibit negligible β_2 and large, negative β_4 (ref. ²⁶). However, the fabrication tolerances that are required for achieving the necessary control of dispersion are very tight. Consequently, PQSs have not been observed experimentally in optical fibres until now.

Here we report the first experimental demonstration of a mode-locked fibre laser that emits PQS pulses. We achieve the necessary laser cavity dispersion conditions by utilizing an intracavity spectral pulse shaper that simultaneously cancels the β_2 and β_3 of the otherwise conventional-fibre cavity, while imparting a strong, negative β_4 . PQSs arise from this lumped quartic dispersion in much the same way that conventional solitons can arise in laser cavities with discrete segments of normal and anomalous dispersion fibre^{29,30}.

Using spectral, temporal and phase-resolved measurements, we demonstrate that the output pulses behave notably differently than any other type of laser pulses: we find that the energy of the emitted pulses (E_{PQS}) is proportional to the third power of the inverse pulse duration, τ^{-3} , which is consistent with theoretical predictions²⁷ and contrasts with conventional solitons, for which $E \propto \tau^{-1}$ (ref. ³¹). We also find that the output spectra exhibit sidebands, which we associate with resonant dispersive waves. Although these sidebands also occur in conventional soliton lasers¹⁴, the sideband frequencies measured here are characteristic for cavities with quartic dispersion and obey a new equation that we present in this work. These results constitute strong evidence for a novel operating regime for mode-locked lasers, opening the way for the generation of high-energy, ultrashort optical pulses that arise from self-phase modulation and higher-order cavity dispersion.

Results

PQS laser cavity. Our laser configuration is shown schematically in Fig. 1a. It is an erbium-doped fibre laser that uses nonlinear

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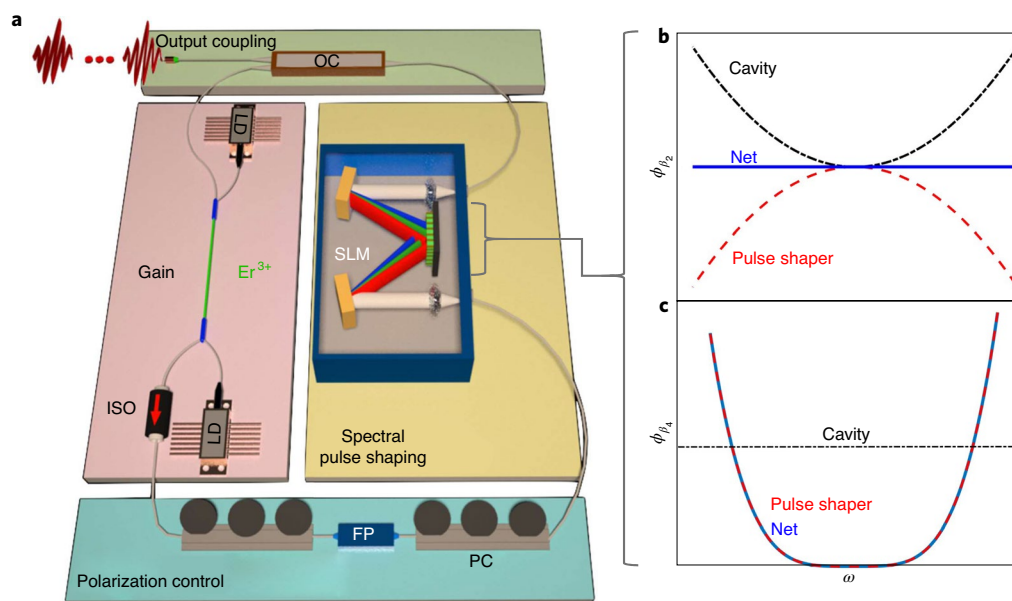


Fig. 1 | The principle of operation of the PQS laser. a, A schematic of the erbium-doped laser cavity with the following components: Er³⁺, erbium-doped fibre; LD, 976 nm laser diode; FP, in-line fibre polarizer; PC, polarization controller; SLM, spatial light modulator; OC, output coupler. **b, c**, Conceptual illustrations of the quadratic (**b**) and quartic (**c**) dispersive phases imparted by the cavity (black dash-dot line) and the spectral pulse shaper (red dashed line), with the net quadratic and quartic phase shown in blue.

polarization evolution for mode locking and incorporates an intra-cavity programmable optical pulse shaper^{32,33}. The total cavity length is approximately $L = 21.4$ m, leading to a fundamental repetition rate of 9.3 MHz. Mode locking is achieved by adjusting the pump power and the polarization controllers. An output coupler is located after the pulse shaper and extracts 50% of the intracavity power. The pulse shaper, based on a spatial light modulator, can produce an arbitrary phase mask and is used to adjust the net cavity dispersion as illustrated in Fig. 1b,c.

The laser operates in the conventional soliton regime when no phase mask is applied, with the intrinsic anomalous group-velocity dispersion of the fibre balancing the self-phase modulation. The output pulses display the well-known hyperbolic secant spectrum centred around 1,563 nm, with a 3.72 nm bandwidth at -3 dB and strong spectral (Kelly) sidebands¹⁴ (Fig. 2a, solid blue curve). The truncated long wavelength edge is due to the finite spectral bandwidth of the pulse shaper. We also perform a set of temporal and phase-resolved measurements using a frequency-resolved electrical gating (FREG) set-up to gain insight into the temporal properties of the output pulses^{25,34}. The retrieved pulse duration of the laser operating in a conventional soliton regime (Fig. 2b, solid blue line) is $\tau = 1.23$ ps.

We then program the pulse shaper to induce a spectral phase profile ϕ that can be written as

$$\phi(\omega) = L \sum_{n=2}^4 \frac{\beta_n \omega^n}{n!}, \quad (1)$$

where ω is the angular frequency. For the results discussed below, $\beta_2 = +21.4 \text{ ps}^2 \text{ km}^{-1}$, $\beta_3 = -0.12 \text{ ps}^3 \text{ km}^{-1}$, and we initially take $\beta_4 = -80 \text{ ps}^4 \text{ km}^{-1}$. The pulse shaper serves two aims: (i) to compensate the second- and third-order dispersion introduced by the optical fibres³⁵ so that the net cavity dispersion of these orders is negligible (Fig. 1b); and (ii) generate a large negative FOD that dominates the propagation (Fig. 1c). The values of the second- and third-order dispersion used as compensation for the single-mode fibre in our set-up were based on those reported in ref. ³⁵. We note that the analysis of our results in Fig. 3b,d confirms that the second- and third-order dispersions in our cavity vanish. The applied FOD

is several orders of magnitude larger than the intrinsic FOD of single-mode fibres ($-80 \text{ ps}^4 \text{ km}^{-1}$ versus $-2.2 \times 10^{-3} \text{ ps}^4 \text{ km}^{-1}$)³⁶. The output spectrum of the laser operating in this regime (shown in Fig. 2e) has a 3.16 nm bandwidth with a shape that differs notably from those of conventional solitons, but is in very good agreement with the theoretically predicted spectral profile of the PQS for the same spectral bandwidth (red dashed line)²⁷. We note in particular the PQS's distinct flatness of the spectral maximum. We also observe several strong, narrowly spaced spectral sidebands; the retrieved pulse corresponding to this spectrum is shown in Fig. 2f (blue solid line). The recovered pulse has a duration of $\tau = 1.74$ ps (full width at half maximum). The retrieved temporal pulse shape is again in good agreement with the theoretically predicted PQS shape for the same pulse duration (red dashed line)²⁷. From this measurement and the spectral bandwidth at -3 dB that is measured from the corresponding spectrum, we calculate a time-bandwidth product of 0.67; this value is larger than the predicted value of 0.53 for transform-limited PQSs²⁷, indicating that the pulses at the laser output are slightly chirped, which is consistent with the recovered phase (green solid line in Fig. 2f). This is not surprising as the output coupler is located immediately after the pulse shaper, which applies the large negative quartic dispersion at a single point in the cavity (see Supplementary Information). Placing the output coupler at a different relative position in the cavity would ensure a reduced time-bandwidth product. Finally, we note that the predicted temporal shape of the PQS exhibits periodic oscillations in the tails that are not observed in the retrieved experimental profile²⁷. This is because the first lobe is expected to appear approximately 28 dB below the pulse's central maximum, which is below the continuous background arising from the dispersive spectral sidebands in our experiments.

We model the laser dynamics using an iterative cavity map^{37,38} in which the propagation through every element is modelled using a generalized nonlinear Schrödinger equation with Kerr nonlinearity and with dispersion up to the fourth order (see Methods). The simulated spectra and intensity profiles corresponding to the two operating regimes are shown in Fig. 2c,d,g,h; they are in excellent agreement with the experimental results (see Supplementary Information) and are consistent with the change in the pulse-shaping mechanism.

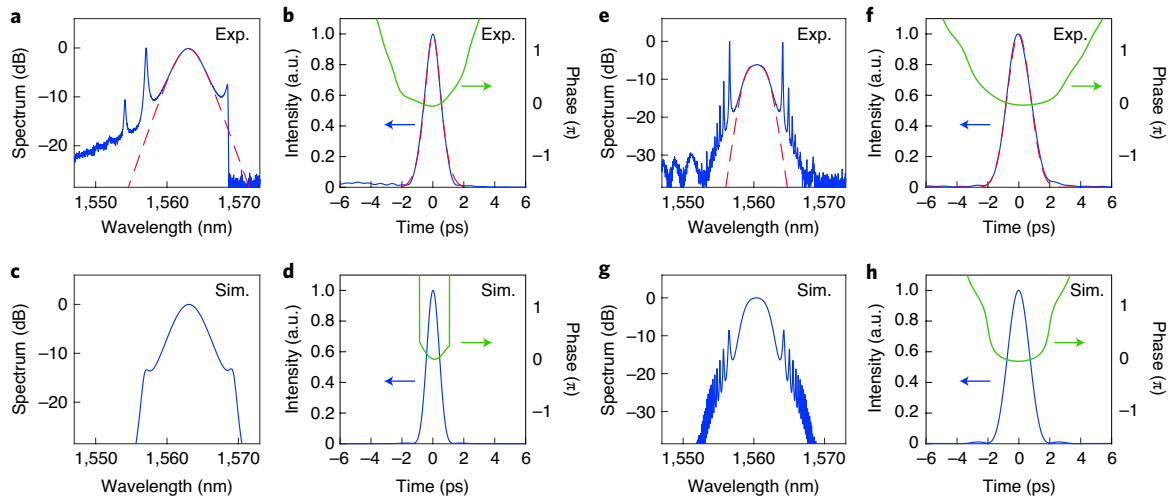


Fig. 2 | Conventional soliton and PQS regimes. **a–h**, Experimental (exp.) and simulated (sim.) output characteristics of the laser in the conventional soliton (**a–d**) and PQS (**e–h**) regimes. **a, e**, Measured output spectra (solid blue curves); red dashed curves show the corresponding soliton and PQS spectral shapes. **b, f**, Retrieved (solid blue curves) and theoretically predicted (red dashed curves) temporal intensity profiles; the solid green curves are the recovered temporal phases. **c, g**, Simulated output spectra. **d, h**, Simulated intensity profiles (blue) and phases (green).

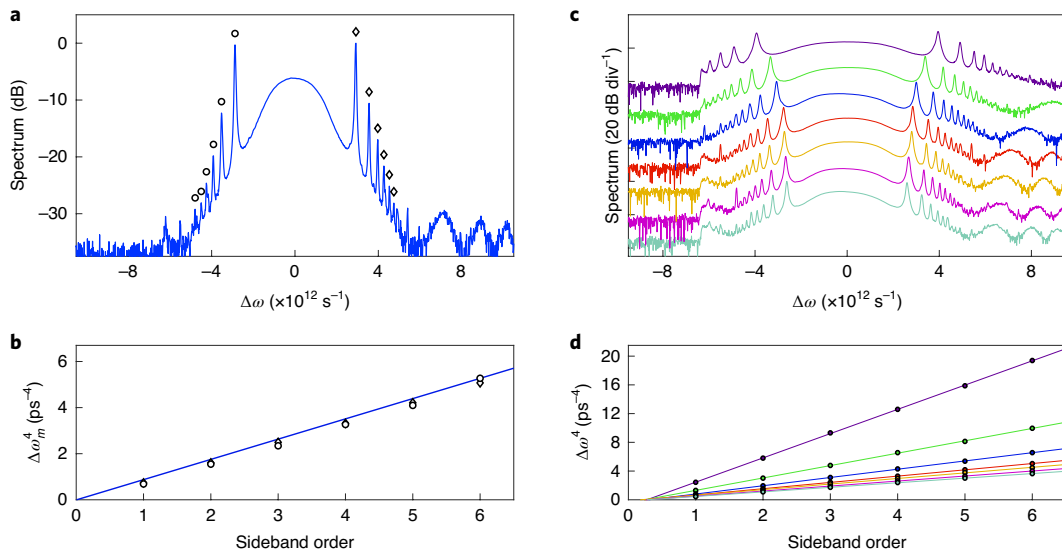


Fig. 3 | Sideband analysis. **a**, A measured spectrum around the pulse central frequency ω_0 . The symbols mark the measured spectral positions of the sidebands for the low (circles) and high (diamonds) frequencies. **b**, The fourth power of the sideband positions as a function of the sideband order; the solid blue line shows the nominal spectral positions of the sidebands that are calculated from equation (2). **c**, Measured optical spectra that correspond to the shortest pulse duration recorded for each value of β_4 . From top to bottom, $\beta_4 = -20 \text{ ps}^4 \text{ km}^{-1}$ (purple), $-40 \text{ ps}^4 \text{ km}^{-1}$ (green), $-60 \text{ ps}^4 \text{ km}^{-1}$ (blue), $-80 \text{ ps}^4 \text{ km}^{-1}$ (red), $-90 \text{ ps}^4 \text{ km}^{-1}$ (yellow), $-100 \text{ ps}^4 \text{ km}^{-1}$ (pink) and $-110 \text{ ps}^4 \text{ km}^{-1}$ (light blue). **d**, The corresponding fourth power of the sideband positions as a function of the sideband order. The solid colour lines correspond to linear fits.

Spectral sidebands analysis. We can assess the quartic nature of the laser cavity dispersion by analysing the spectral positions of the dispersive waves in Fig. 2e. Similar to conventional solitons, these dispersive waves arise from perturbations when the soliton propagates through the cavity; the spectral positions of these peaks provide information about the cavity dispersion³⁹. The dispersive waves are generated every round trip and interfere constructively when $\beta_{\text{PQS}} - \beta_{\text{lin}} = 2m\pi/L$, where β_{PQS} and β_{lin} are the propagation constants of the soliton and the dispersive wave, respectively, and m is an integer. When this condition is satisfied, the dispersive waves that are generated in consecutive passages through the cavity interfere constructively, leading to narrow spectral peaks¹⁴.

For a linear wave propagating in a quartic dispersion cavity, $\beta_{\text{lin}} = -|\beta_4|(\omega - \omega_0)^4/24$, whereas the PQS experiences a constant dispersion across its entire bandwidth of $\beta_{\text{PQS}} = K|\beta_4|/\tau^4$, where the constant $K = 1.67$ (ref. 27). Using the argument outlined in the previous paragraph, it is straightforward to show that the m th-order spectral resonances (ω_m) satisfy

$$\omega_m = \pm \frac{1}{\tau} \left(\frac{48m\pi\tau^4}{|\beta_4|L} - 24K \right)^{1/4} \quad (2)$$

This equation shows that the fourth power of the sideband frequency offsets are equally spaced by $48\pi/(\beta_4 L)$. In Fig. 3a we show

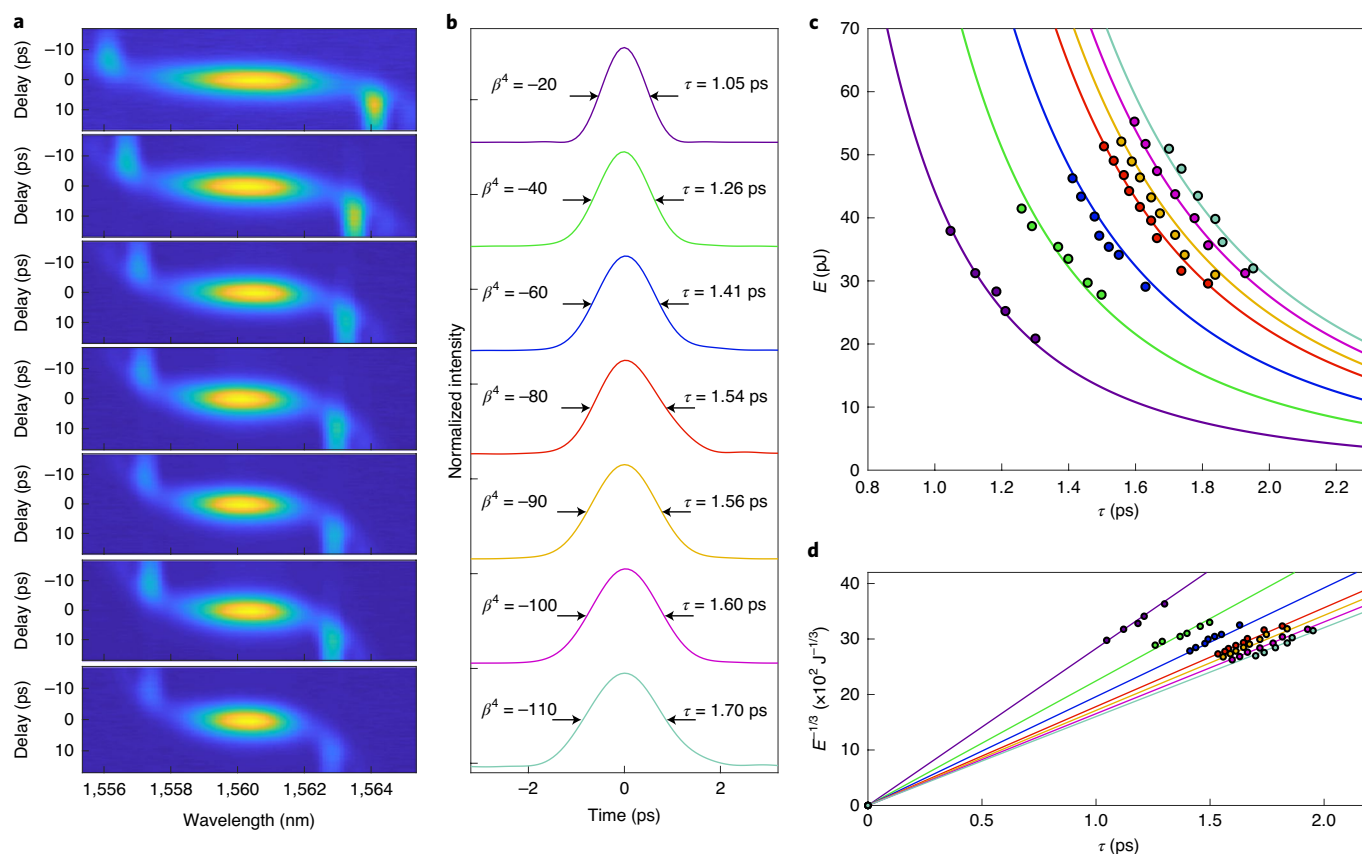


Fig. 4 | Measured energy-width scaling properties of the emitted PQS pulses for different values of applied quartic dispersion. a, b, Measured spectrograms of the shortest pulse durations recorded for each value of β_4 (**a**) and the corresponding retrieved temporal intensity profiles (**b**); the values of applied β_4 are indicated on the left of **b** (in $\text{ps}^4 \text{km}^{-1}$). From top to bottom, $\beta_4 = -20 \text{ ps}^4 \text{km}^{-1}$ (purple), $-40 \text{ ps}^4 \text{km}^{-1}$ (green), $-60 \text{ ps}^4 \text{km}^{-1}$ (blue), $-80 \text{ ps}^4 \text{km}^{-1}$ (red), $-90 \text{ ps}^4 \text{km}^{-1}$ (yellow), $-100 \text{ ps}^4 \text{km}^{-1}$ (pink) and $-110 \text{ ps}^4 \text{km}^{-1}$ (light blue). **c,** Measured pulse energy (E) versus pulse duration (circles). The solid curves are $\propto \beta_4 \tau^{-3}$, as follows from equation (3). **d,** $E^{-1/3}$ versus pulse duration shown over a wider range.

the optical spectrum of the PQS centred around ω_0 and we mark the measured spectral positions of the sidebands for the low (circles) and high (diamonds) frequencies of the spectrum. Figure 3b shows the fourth power of these sideband positions ($\Delta\omega^4$) versus the sideband order, confirming that they follow a linear relation, consistent with equation (2). The spacing determined from the measured spectral position of the sidebands is fixed at 0.90 ps^{-4} and 0.867 ps^{-4} for the low (circles) and high (diamonds) frequencies, respectively. These results are in excellent agreement with the theoretical value of 0.88 ps^{-4} following from equation (2). The small discrepancies between the calculated and theoretical values could be due to the limited resolution of the measurement of the position of the sidebands or residual, uncompensated quadratic and cubic dispersion.

Although the results discussed thus far were taken at a fixed $\beta_4 = -80 \text{ ps}^4 \text{km}^{-1}$, in Fig. 3c,d we show similar measurements for seven different values of the quartic dispersion; namely, $\beta_4 = -20, -40, -60, -80, -90, -100$ and $-110 \text{ ps}^4 \text{km}^{-1}$. Figure 3c shows measured spectra for the shortest pulses that we generated at each of these β_4 values (see Fig. 4b). The similarity of these spectra implies that the pulse-shaping mechanism is unchanged for all of the β_4 values considered. Similarly, in Fig. 3d we show an analysis of the positions of the sidebands. As all of the $\Delta\omega^4$ values lie on straight lines, we confirm that the dispersion remains quartic for all values of β_4 . The presence of the last term in the brackets in equation (2) implies that the straight lines do not go through the origin or even through a common point.

In Fig. 4a we show spectrograms of the shortest pulses measured for each value of β_4 , corresponding to the spectra shown in

Fig. 3c. The corresponding retrieved temporal intensity profiles are shown in Fig. 4b. Similar to conventional soliton lasers, reducing the net-cavity dispersion allows for generation of shorter optical pulses^{30,40,41}. The vertical streaks at short and long wavelengths in Fig. 4a are the first sidebands on either side of the pulse spectrum, as discussed earlier. On the short wavelength side they precede the pulse in time, whereas on the long wavelength side they follow it, as per the requirements for negative quartic dispersion.

Energy-width scaling. Finally, we report the energy-width scaling of the emitted PQSs. Although the pulse energy is inversely proportional to the pulse duration for conventional solitons, recent theoretical and numerical studies show that for a PQS the pulse energy is given by²⁷

$$E_{\text{PQS}} = \frac{2.87|\beta_4|}{\gamma\tau^3} \quad (3)$$

where γ is the average cavity nonlinear parameter. At each value of the quartic dispersion we measure the spectral bandwidth of the output pulses for different pulse energies by adjusting the pump power, and we deduct the portion of the energy in the spectral sidebands by integrating the measured optical spectrum. The corresponding pulse durations are then calculated using the time-bandwidth product of 0.67 determined above, which we find to be constant for the entire range of parameters that we consider. The results of this procedure are summarized in Fig. 4c,d. The circles in Fig. 4c show the measured pulse energies versus the pulse duration (τ) for different values of quartic dispersion. The results in Fig. 4c are in excellent

agreement with equation (3) once we account for the output coupling and the variations of the pulse parameters within the cavity (see Methods). This shows that $E \propto \tau^{-3}$ for fixed β_4 and $E \propto \beta_4$ for fixed τ , consistent with equation (3).

Figure 4d shows the same data as Fig. 4c, but has $E^{-1/3}$ on the vertical axis. Plotted in this way, the data should form a fan of straight lines, with each rib corresponding to a particular value of β_4 , going through the origin. The consistency between the measured data and the prediction from equation (3) is conclusive evidence of the unique scaling properties of the pulses emitted by our laser.

A second set of measurements (not shown here) is performed with a different output coupler, extracting 10% of the intracavity power, and we obtain similar results. We also perform a similar measurement (see Supplementary Information) for the laser operating in the conventional soliton regime (as seen in Fig. 2a) and find that the output pulse energy follows the well-known relation $E \propto \tau^{-1}$ (ref. 31). These results confirm that PQS pulses follow a different energy-width scaling relation and that they could outperform conventional soliton for short pulse durations, as has been suggested by previous studies^{27,42}.

Discussion

We provide the first experimental realization of a PQS laser by harnessing the interaction between FOD and nonlinearity in a fibre laser cavity. We have provided experimental evidence of a novel pulse spectrum—with a temporal-shape, energy-width scaling and spectral-sideband behaviour—with respect to any types of laser pulses produced so far.

The necessary FOD-dominating dispersion profile was achieved by incorporating a reconfigurable spectral pulse shaper into the laser cavity. This innovation is crucial to leverage the effects of higher-order dispersion in ultrafast lasers without the need for challenging dispersion engineering of the waveguide or fibre.

The range of pulse energies achievable with our current set-up is limited by the spectral and delay range of the spectral pulse shaper, this limitation is in no way fundamental and it could be overcome by using a more broadband pulse shaper with a larger dynamic range. In the future, fibre platforms with appropriate dispersion²⁶ or fixed phase masks could substitute the pulse shaper to build a new class of mode-locked laser that emits ultrashort high-energy soliton pulses.

Alternatively, microresonator geometries with dominant negative FOD have recently been theoretically studied²⁸. This highlights the potential of on-chip PQSs for the generation of broadband coherent frequency combs with small line-to-line power variations and may lead to more power-efficient wavelength-division multiplexing for telecommunication systems⁴³. On-chip high-energy pulses could also be achieved based on PQS effects and support the development of new nonlinear application in integrated photonics.

Finally, the simple laser cavity configuration presented in this work offers an exceedingly flexible testbed for the generation and the study of optical pulses arising from the interplay of Kerr nonlinearity and hybrid dispersion—combination of different orders of dispersion, including orders higher than fourth. This provides new degrees of freedom to optically control the shape and the energy scaling of optical pulses. This could not only have impact on ultrafast lasers, but also on other areas in which these features are crucial such as on-chip frequency combs, supercontinuum generation, and advanced modulation formats for communications.

Online content

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Methods

Numerical simulation model. Numerical simulations are based on the nonlinear Schrödinger equation:

$$\frac{\partial A}{\partial z} + iD\left(i\frac{\partial}{\partial T}\right)A = \frac{g}{2}A + i\gamma A|A|^2 \quad (4)$$

Here, $A = A(z, T)$ is the slowly varying amplitude of the pulse envelope, z is the propagation coordinate, i is the imaginary unit, T is the pulse local time, g is the gain coefficient and γ is the nonlinear parameter given by $\gamma = n_2\omega_0/(cA_{\text{eff}})$, in which n_2 is the nonlinear refractive index, ω_0 the central frequency, c the speed of light in vacuum and A_{eff} the effective mode area. The dispersion operator is defined as:

$$D\left(i\frac{\partial}{\partial T}\right) = \sum_k \frac{\beta_k}{k!} \left(i\frac{\partial}{\partial T}\right)^k, \quad (5)$$

for $k=2, 3, 4$; β_k is the k th order of dispersion. The gain g in the doped fibre section is calculated using:

$$g = \frac{g_0}{1 + E(z)/E_{\text{sat}}} \quad (6)$$

where $g_0 = 3.45$ and is the small-signal gain (corresponding to 30 dB in power and is non-zero only in the doped-fibre section), $E(z) = \int |A(z, T)|^2$ is the pulse energy and E_{sat} is the saturation energy, which is adjusted to simulate changing the pump power. We multiply $g(z)$ with a Lorentzian profile of 50 nm width to form the finite gain bandwidth $g(z, \omega_0)$. The saturable absorber is modelled by a transfer function that describes its transmittance

$$T(\tau) = 1 - \frac{q_0}{1 + P(\tau)/P_0} \quad (7)$$

where q_0 is the unsaturated loss of the saturable absorber, $P(\tau) = |A(z, \tau)|^2$ is the instantaneous pulse power and P_0 is the saturation power. The spectral pulse shaper is modelled by multiplying the electric field by a phase following the expression in equation (1) in the spectral domain. The insertion losses (~5.6 dB) of the spectral pulse shaper are also taken into account in the simulations. Our numerical model is solved with a standard symmetric split-step Fourier method algorithm. The dispersion and gain contributions are calculated in the frequency domain, whereas the nonlinear term is calculated in the time domain. For our simulations we have used an initial field composed of Gaussian random noise multiplied by a sech shape in the time domain. The same stable solutions are reached for different initial noise fields.

The parameters used in our numerical simulations are the same as their experimental values. The erbium-doped fibre is 1.5 m long, with a mode-field diameter of 9.5 μm , numerical aperture of 0.13 and $\gamma = 0.0016 \text{ W}^{-1} \text{ m}^{-1}$ at 1,560 nm. The rest of the cavity is built from SMF-28 fibre. SMF-28 has a mode-field diameter of 10.4 μm , numerical aperture of 0.14, $\gamma = 0.0013 \text{ W}^{-1} \text{ m}^{-1}$ at 1,560 nm. The dispersion coefficients are, $\beta_2 = -21.4 \text{ ps}^2 \text{ km}^{-1}$, $\beta_3 = 0.12 \text{ ps}^3 \text{ km}^{-1}$ and $\beta_4 = -0.0022 \text{ ps}^4 \text{ km}^{-1}$. As the length of erbium-doped fibre is much shorter than the total cavity length and the mode-field diameter and numerical aperture are close to that of SMF-28, we assumed similar dispersion coefficients. For the simulated results shown in Fig. 2, $q_0 = 0.7$ and $P_0 = 200 \text{ W}$. The saturation energy was set at $E_{\text{sat}} = 90 \text{ pJ}$ and $E_{\text{sat}} = 100 \text{ pJ}$, for the conventional and PQSs, respectively.

Mode locking. Light (980 nm) from two laser diodes is delivered to the cavity through two 980/1,550 nm wavelength division multiplexers. An optical isolator is used to ensure unidirectional pulse propagation in the cavity. Nonlinear polarization evolution is implemented using a set of two fibre polarization

controllers and a fibre polarizer, which act as an artificial saturable absorber.

At a pump power of 370 mW, the laser is self-starting and multipulsing for both mode-locking regimes after adjusting the polarization controller. Single pulsing is then achieved by decreasing the pump power to 155 mW, at which the average output power is 120 μW and 130 μW for the conventional and PQS regimes, respectively. The pump power could be continuously decreased to 120 mW, leading to an average output power of 65 μW . The pulse energies are calculated after taking the constant output coupling and pulse-shaper losses into account.

Phase-resolved characterization method. The output pulses were input into the FREG apparatus. The pulses were split into two branches by a 70/30 fibre-coupler; 30% of the output power was sent to a branch with a variable delay before being detected by a fast photodiode and transferred to the electrical domain. This electrical signal drove a Mach-Zehnder modulator that gated the optical pulses from the 70% branch of the fibre-coupler. Using an optical spectrum analyser, we measured the spectra as a function of the delay to generate a series of optical spectrograms, which were measured over a 15 nm spectral bandwidth with a $\Delta\lambda = 0.03 \text{ nm}$ resolution, and the temporal delay was scanned over 40 ps with a 80 fs temporal step.

We then deconvolved the spectrograms with a blind deconvolution numerical algorithm (512×512 grid; retrieval errors < 0.005) to retrieve the pulse intensity and the phase in the temporal domain⁴⁴. We verified the validity of the retrieved pulses by taking their Fourier transform and checked these matched with the output spectra measured with the optical spectrum analyser.

Data availability

The data that support the plots in this paper and other findings of this study are available from the corresponding author on reasonable request.

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Author contributions

A.B.-R., C.M.d.S. and D.D.H. conceived the idea of the PQS laser. A.F.J.R., D.D.H. and A.B.-R. designed the experiment. A.F.J.R. performed the experiments and the numerical simulations. K.K.K.T. and C.M.d.S. carried out the theoretical analysis. C.M.d.S. and A.B.-R. supervised the overall project. All the authors contributed to the interpretation of data and wrote the manuscript.

Competing interests

A.F.J.R., D.D.H., C.M.d.S. and A.B.-R. have submitted a provisional patent application based on the ideas presented in this work.

Additional information

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